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Aim : Aim: Energy-Aware MAC Protocol for WSN

Implement duty-cycling MAC protocol and evaluate energy conservation in sensor networks.

```
# =====  
# Energy-Aware MAC Protocol for WSN  
# =====
```

```
set ns [new Simulator]
```

```
# Trace Files
```

```
set tr [open energy.tr w]
```

```
$ns trace-all $tr
```

```
set nf [open energy.nam w]
```

```
$ns namtrace-all-wireless $nf 500 500
```

```
# Topography
```

```
set topo [new Topography]
```

```
$topo load_flatgrid 500 500
```

```
create-god 5
```

```
# Channel
```

```
set chan [new Channel/WirelessChannel]
```

```
# Node Configuration
```

```
$ns node-config \
```

```
-adhocRouting DSDV \
```

```
-llType LL \
```

```
-macType Mac/802_11 \
```

```
-ifqType Queue/DropTail/PriQueue \
```

```
-ifqLen 50 \
```

```
-antType Antenna/OmniAntenna \
```

```
-propType Propagation/TwoRayGround \
```

```
-phyType Phy/WirelessPhy \
```

```
-channel $chan \
```

```
-topoInstance $topo \
```

```
-agentTrace ON \
```

```
-routerTrace ON \
```

```
-macTrace ON \
```

```
-movementTrace ON \
```

```
-energyModel EnergyModel \
```

```
-initialEnergy 10 \
```

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```
-txPower 2.0 \  
-rxPower 1.0 \  
-idlePower 0.5
```

```
# Create Nodes  
for {set i 0} {$i < 5} {incr i} {  
    set node_($i) [$ns node]  
}
```

```
# Node Positions  
$node_(0) set X_ 50  
$node_(0) set Y_ 250  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 150  
$node_(1) set Y_ 250  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 250  
$node_(2) set Y_ 250  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 350  
$node_(3) set Y_ 250  
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 450  
$node_(4) set Y_ 250  
$node_(4) set Z_ 0
```

```
for {set i 0} {$i < 5} {incr i} {  
    $ns initial_node_pos node_($i) 20  
}
```

```
# UDP Source  
set udp0 [new Agent/UDP]  
$ns attach-agent $node_(0) $udp0
```

```
# UDP Sink  
set null0 [new Agent/Null]  
$ns attach-agent $node_(4) $null0
```

```
$ns connect $udp0 $null0
```

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```
# CBR Traffic
set cbr0 [new Application/Traffic/CBR]

$cbr0 set packetSize_ 1000
$cbr0 set interval_ 0.02
$cbr0 attach-agent $udp0

# Start Traffic
nsat1.0 "cbr0 start"
nsat20.0 "cbr0 stop"

# Finish Procedure
proc finish {} {
    global ns tr nf node_

    puts "\n===== ENERGY REPORT ====="

    for {set i 0} {$i < 5} {incr i} {
        puts "Node i RemainingEnergy = [node_($i) energy] J"
    }

    close $tr
    close $nf

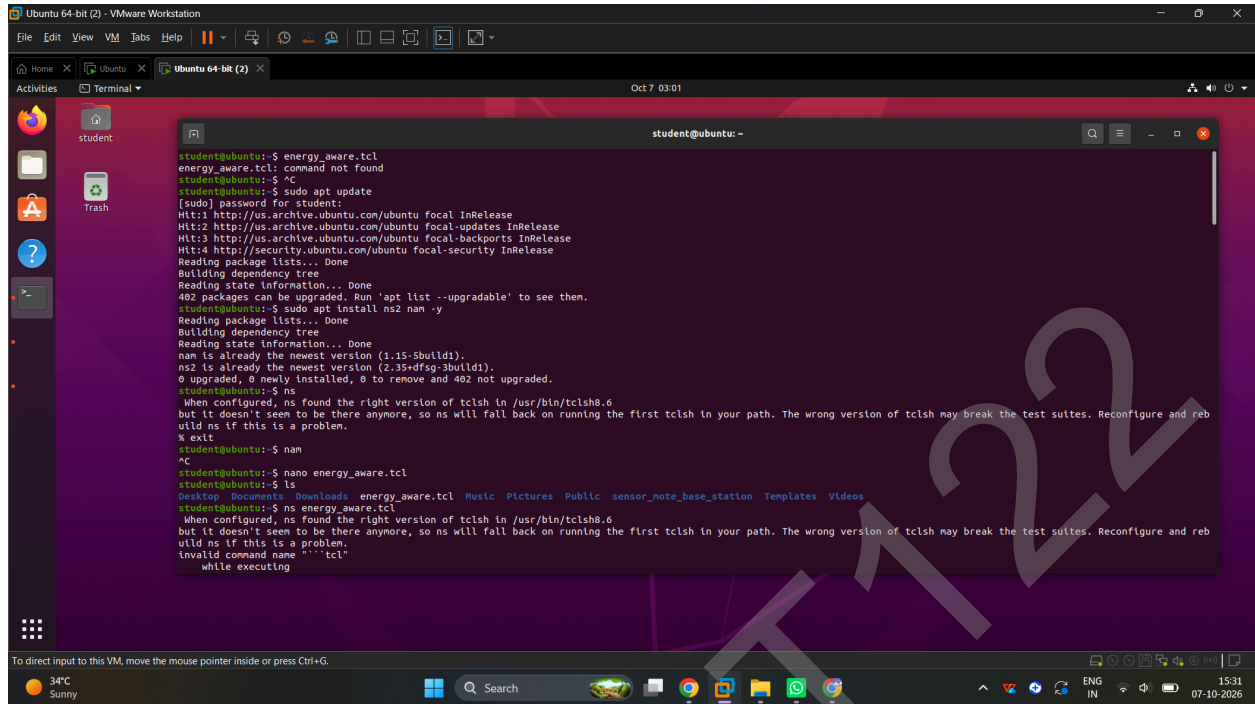
    exec nam energy.nam &
    exit 0
}

$ns at 25.0 "finish"

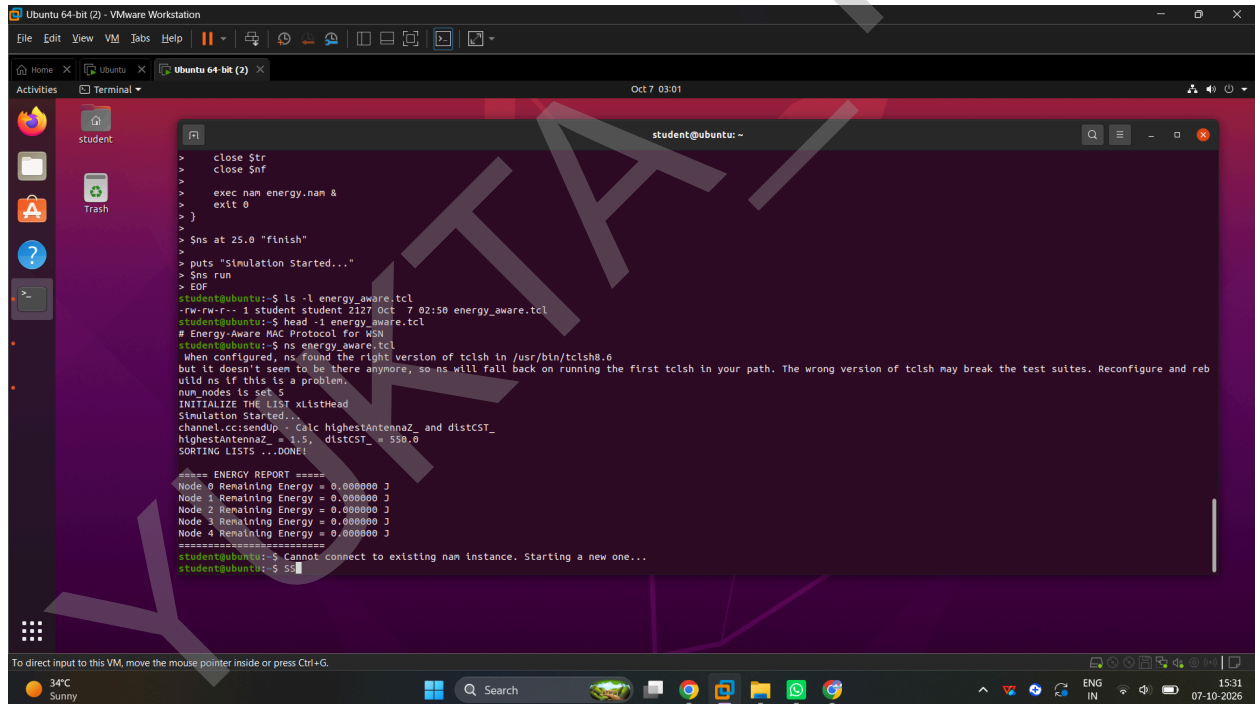
puts "Simulation Started..."

$ns run
```

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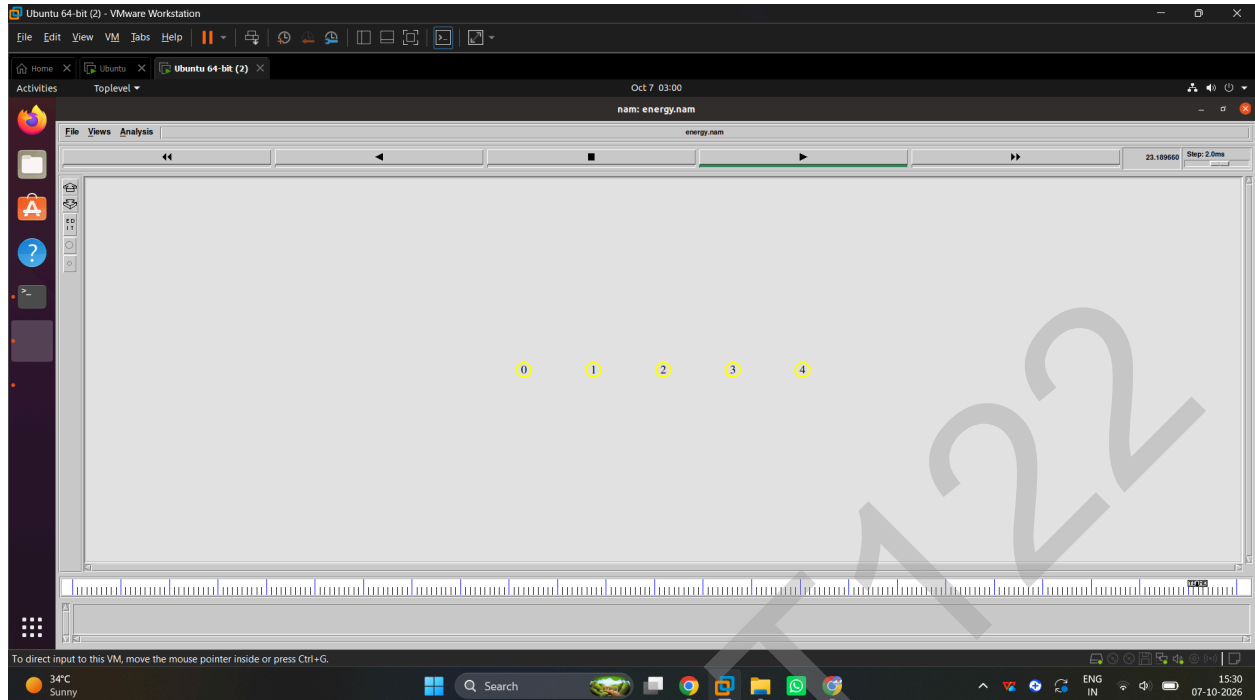


```
student@ubuntu:~$ energy_aware.tcl
energy_aware.tcl: command not found
student@ubuntu:~$ ^C
student@ubuntu:~$ sudo apt update
[sudo] password for student:
Hit:1 http://us.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
402 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@ubuntu:~$ sudo apt install ns2 nam -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
ns2 is already the newest version (1:15.5build1).
ns2 is already the newest version (2:35dfsg-3build1).
0 upgraded, 0 newly installed, 0 to remove and 402 not upgraded.
student@ubuntu:~$ ns
When configured, ns found the right version of tcsh in /usr/bin/tcsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tcsh in your path. The wrong version of tcsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
% exit
student@ubuntu:~$ nam
^C
student@ubuntu:~$ nano energy_aware.tcl
student@ubuntu:~$ ls
Desktop Documents Downloads energy_aware.tcl Music Pictures Public sensor_note_base_station Templates Videos
student@ubuntu:~$ ns energy_aware.tcl
When configured, ns found the right version of tcsh in /usr/bin/tcsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tcsh in your path. The wrong version of tcsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
invalid command name ""tcsh""
while executing
```



```
> close $tr
> close $nf
> exec nam energy.nam &
> exit 0
> }
>
> $ns at 25.0 "finish"
> puts "Simulation Started..."
> $ns run
> EOF
student@ubuntu:~$ ls -l energy_aware.tcl
-rw-rw-r-- 1 student student 2127 Oct  7 02:50 energy_aware.tcl
student@ubuntu:~$ head -1 energy_aware.tcl
# Energy-Aware MAC Protocol for WSN
student@ubuntu:~$ ns energy_aware.tcl
When configured, ns found the right version of tcsh in /usr/bin/tcsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tcsh in your path. The wrong version of tcsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
num_nodes is set 5
INITIALIZE THE LIST xListHead
Simulation Started...
channel.cc:sendup - Calc HighestAntennaZ_and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
===== ENERGY REPORT =====
Node 0 Remaining Energy = 0.000000 J
Node 1 Remaining Energy = 0.000000 J
Node 2 Remaining Energy = 0.000000 J
Node 3 Remaining Energy = 0.000000 J
Node 4 Remaining Energy = 0.000000 J
=====
student@ubuntu:~$ Cannot connect to existing nam instance. Starting a new one...
student@ubuntu:~$ ss
```

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